

11(a)	conduction band and valence band overlap	B1
	number (density) of charge carriers does not vary	B1
	increase in temperature gives rise to <u>increased</u> lattice vibrations	B1
	(lattice) vibrations hinder movement of charge carriers so resistance increases	B1
11(b)	$mv = h / \lambda$	C1
	$v = (6.63 \times 10^{-34}) / [(2.6 \times 10^{-11}) \times (9.11 \times 10^{-31})]$ (= $2.80 \times 10^7 \text{ m s}^{-1}$)	C1
	$qV = \frac{1}{2}mv^2$	C1
	$V = [9.11 \times 10^{-31} \times (2.80 \times 10^7)^2] / [2 \times 1.60 \times 10^{-19}]$ $= 2.2 \times 10^3 \text{ V}$	A1